

Digital Image Processing

Lecture # 4 **Spatial Enhancement-II**

Point Processing Transformations

- ◆ There are many different kinds of grey level transformations

- ◆ Three of the most common are shown here

- Linear

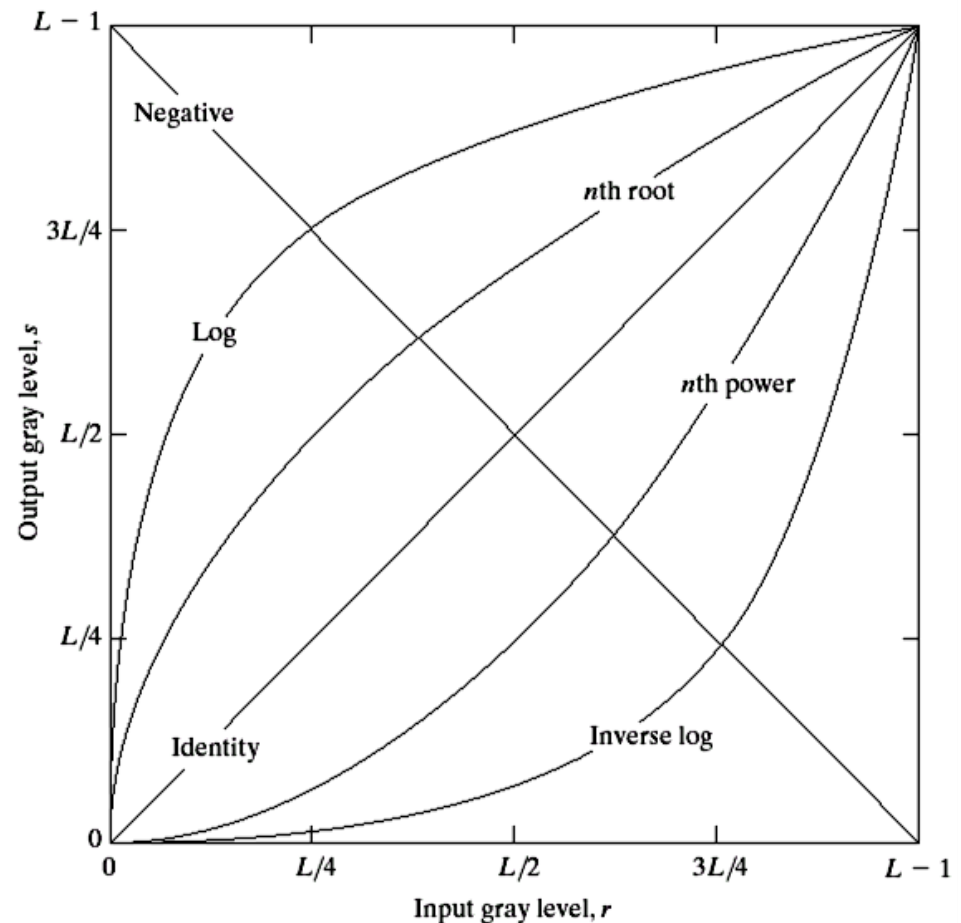
- Negative/Identity

- Logarithmic

- Log/Inverse log

- Power law

- n^{th} power/ n^{th} root



Point Processing Example: Negative Images

- ◆ Reverses the gray level order
- ◆ For L gray levels, the transformation has the form:

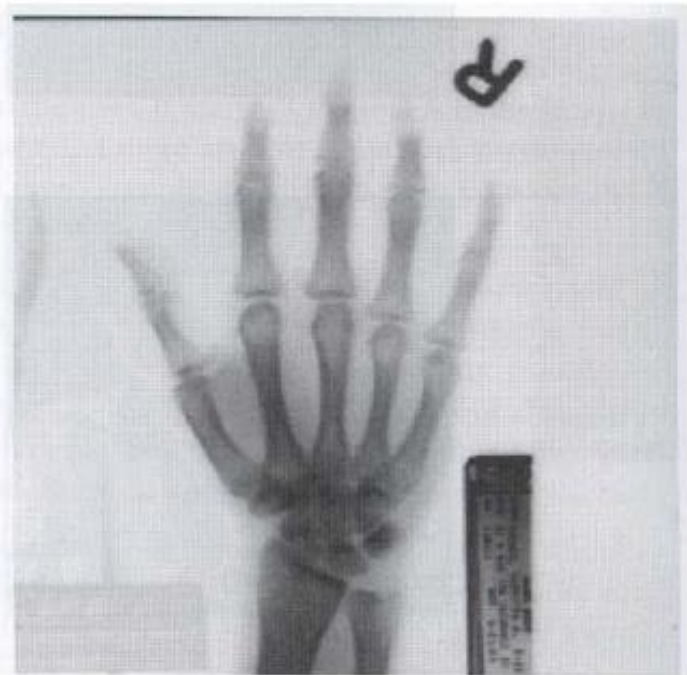
$$s = (L - 1) - r$$

- ◆ Negative images are useful for enhancing white or grey detail embedded in dark regions of an image

Point Processing Example: Negative Images



Input image (X-ray image)



Output image (negative)

Logarithmic Transformations

- ◆ The general form of the log transformation is

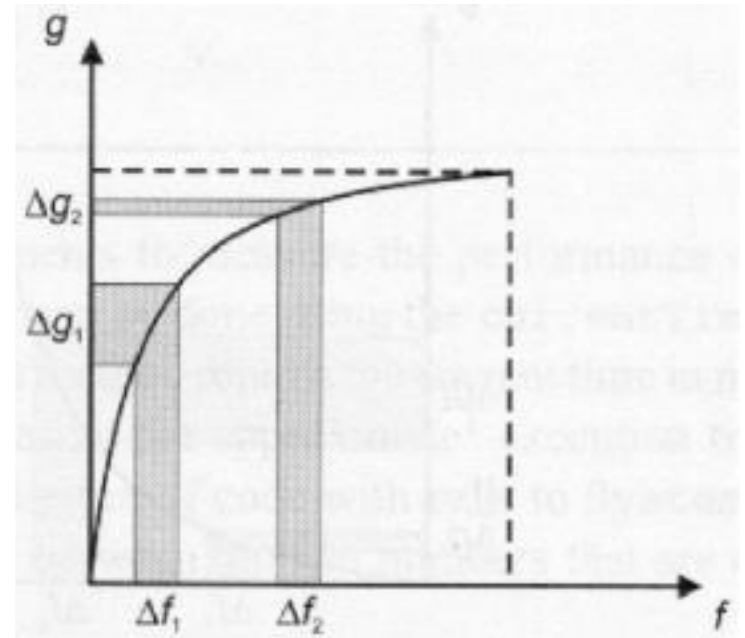
$$s = c \times \log(1 + r)$$

- ◆ The log transformation maps a narrow range of low input grey level values into a wider range of output values
- ◆ The inverse log transformation performs the opposite transformation

Logarithmic Transformations

◆ Properties

- For lower amplitudes of input image the range of gray levels is expanded
- For higher amplitudes of input image the range of gray levels is compressed

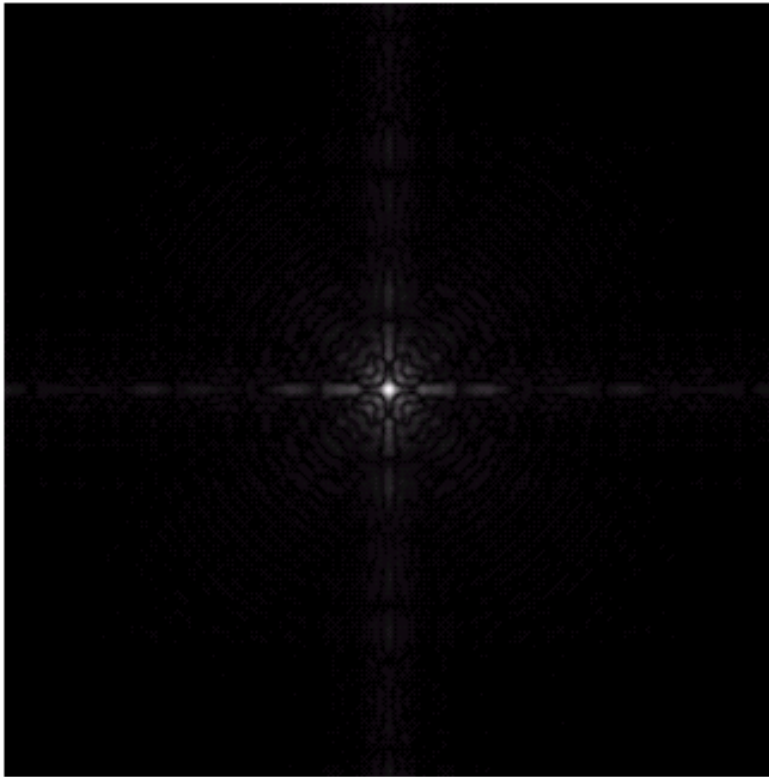


Logarithmic Transformations

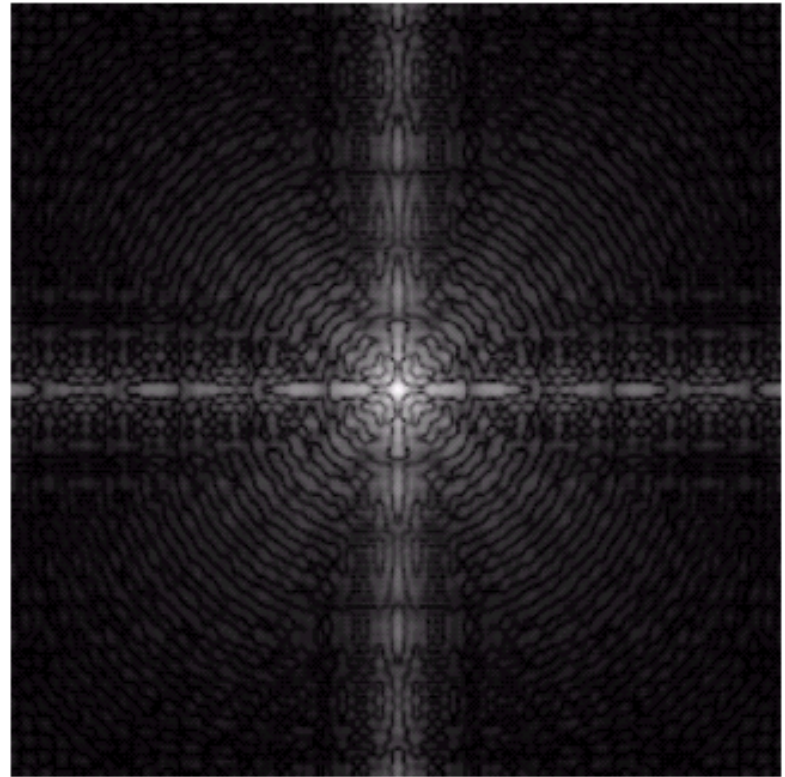
◆ Application

- This transformation is suitable for the case when the dynamic range of a processed image far exceeds the capability of the display device (e.g. display of the Fourier spectrum of an image)
- Also called “dynamic-range compression / expansion”

Logarithmic Transformations



Fourier spectrum: image values ranging from 0 to 1.5×10^6



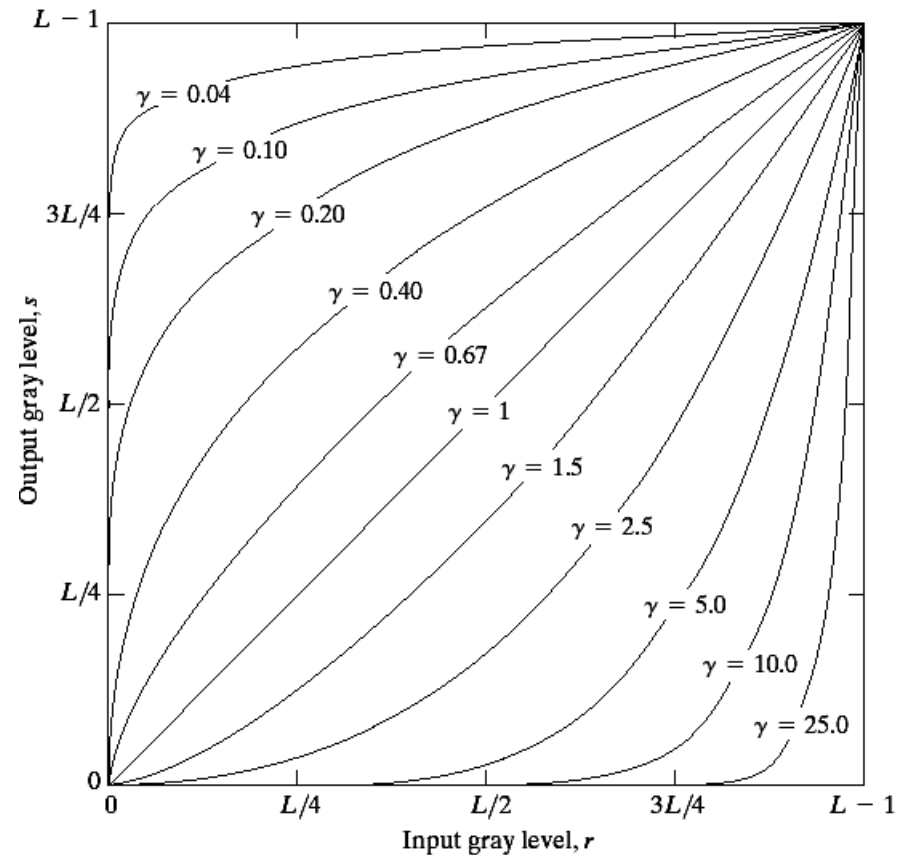
The result of log transformation with $c = 1$

Power Law Transformations

- ◆ Power law transformations have the following form

$$s = c \times r^\gamma$$

- ◆ Map a narrow range of dark input values into a wider range of output values or vice versa
- ◆ Varying γ gives a whole family of curves



Power Law Transformations

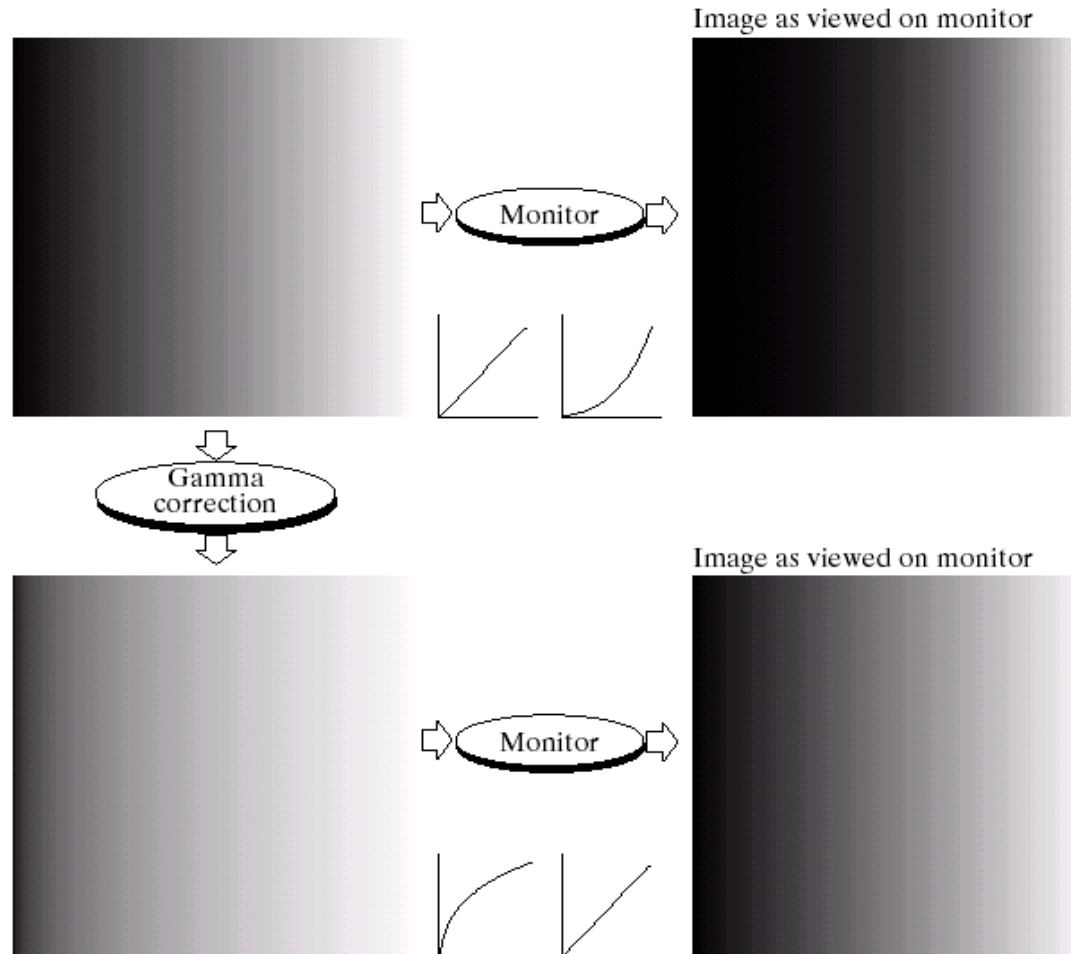
- ◆ For $\gamma < 1$: Expands values of dark pixels, compress values of brighter pixels
- ◆ For $\gamma > 1$: Compresses values of dark pixels, expand values of brighter pixels
- ◆ If $\gamma=1$ & $c=1$: Identity transformation ($s = r$)
- ◆ A variety of devices (image capture, printing, display) respond according to a power law and need to be corrected
- ◆ **Gamma (γ) correction**
The process used to correct the power-law response phenomena

Power Law Transformations: Gamma Correction

a b
c d

FIGURE 3.7

(a) Linear-wedge gray-scale image.
(b) Response of monitor to linear wedge.
(c) Gamma-corrected wedge.
(d) Output of monitor.



Power Law Transformations

Contrast Enhancement

The images to the right show a magnetic resonance (MR) image of a fractured human spine



Power Law Transformations

Contrast Enhancement

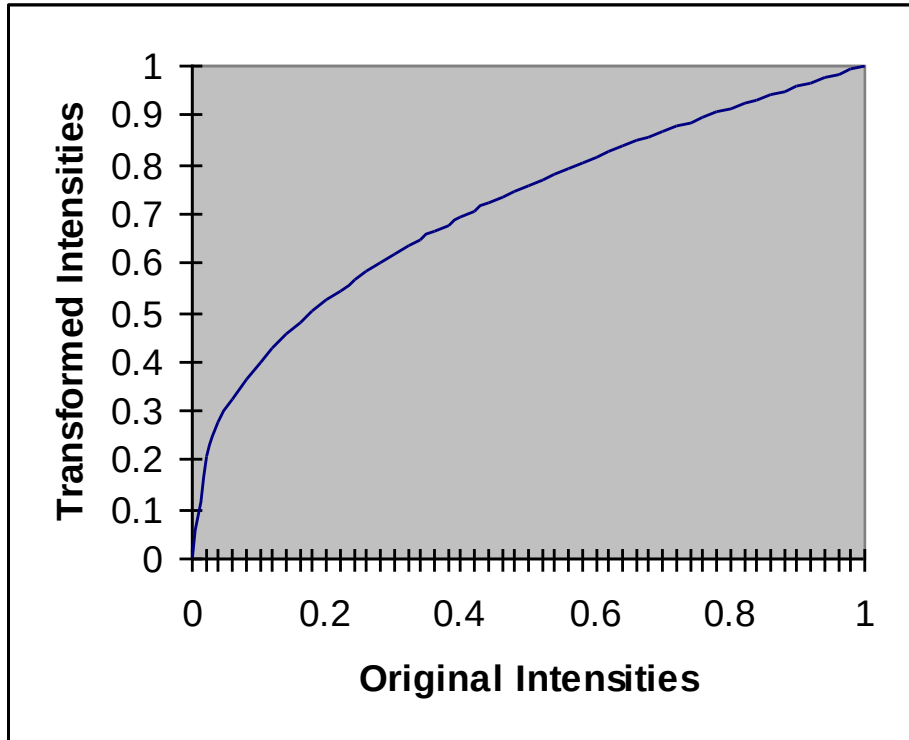
$$\gamma = 0.6$$



Power Law Transformations

Contrast Enhancement

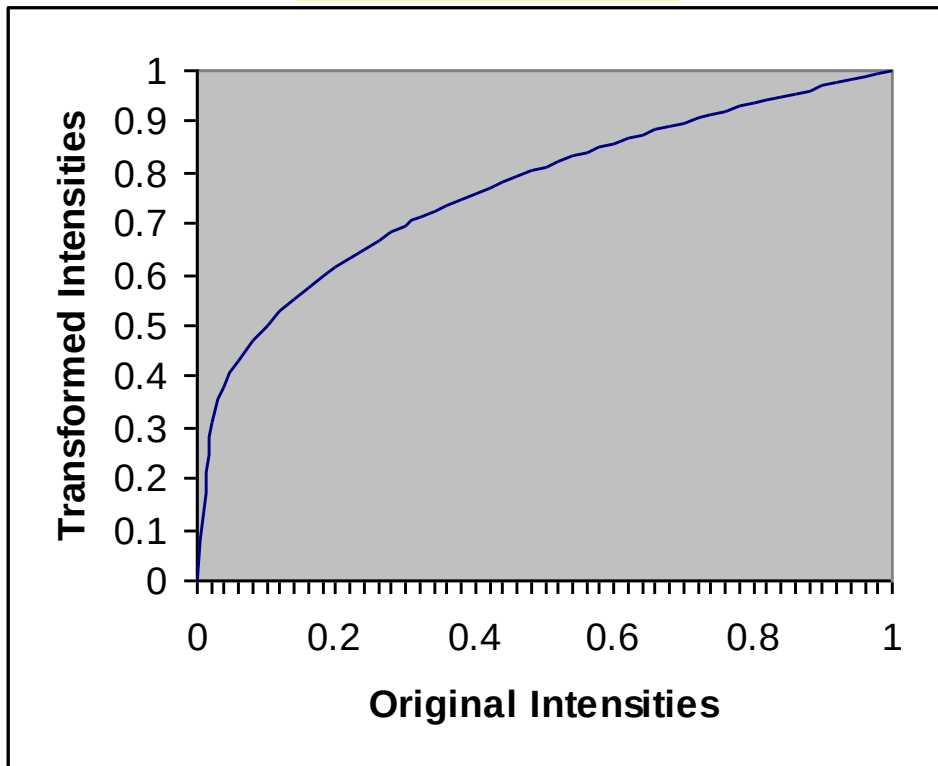
$$\gamma = 0.4$$



Power Law Transformations

Contrast Enhancement

$$\gamma = 0.3$$



Power Law Transformations

Contrast Enhancement



MR image of
fractured human spine



Result after
Power law
transformation

$c = 1, \gamma = 0.6$



Result after
Power law
transformation

$c = 1, \gamma = 0.4$



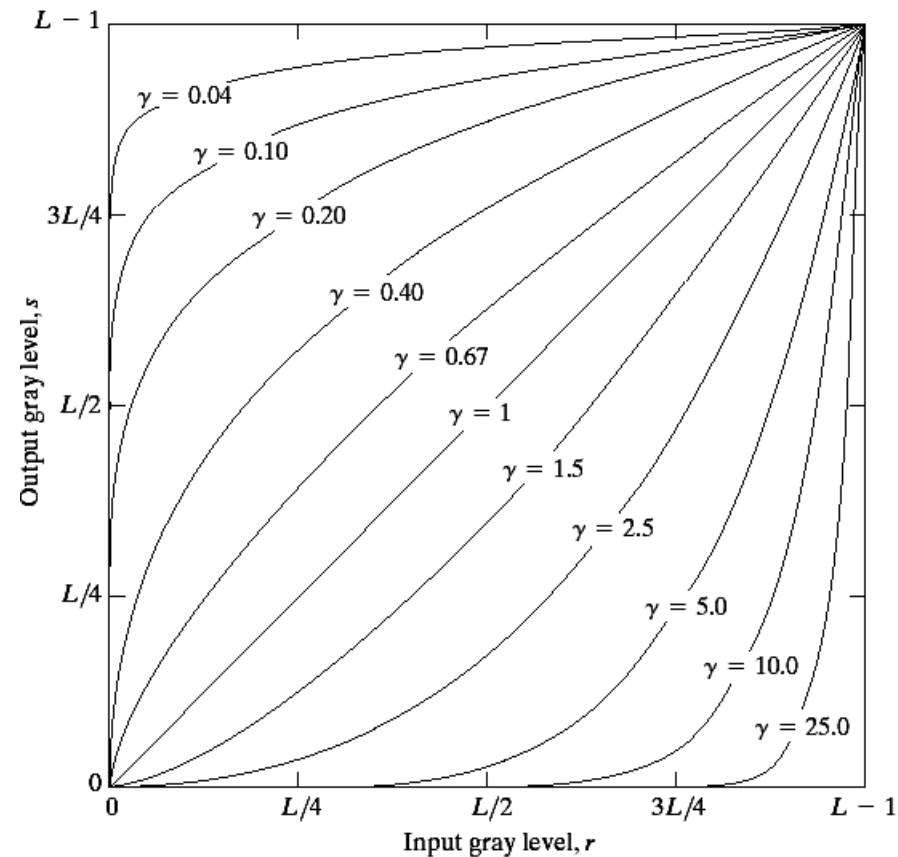
Result after
Power law
transformation

$c = 1, \gamma = 0.3$

Power Law Transformations

Contrast Enhancement

When the γ is reduced too much, the image begins to reduce contrast to the point where the image started to have very slight “wash-out” look.



Power Law Transformations Contrast Enhancement

Image has a washed-out appearance – needs $\gamma > 1$



Image Enhancement

Aerial
Image



Result of
Power law
transformation
 $c = 1, \gamma = 3.0$
(suitable)



Result of
Power law
transformation
 $c = 1, \gamma = 4.0$
(suitable)



Result of
Power law
transformation
 $c = 1, \gamma = 5.0$
(high contrast,
some regions are
too dark)



Histogram of a Grayscale Image

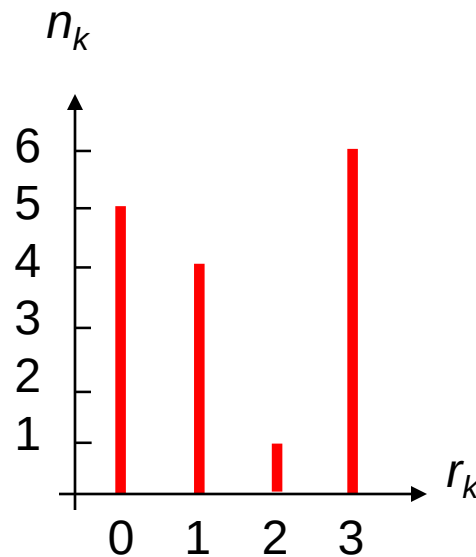
Histogram of a Grayscale Image

- ◆ Let I be a 1-band (grayscale) image.
- ◆ $I(r,c)$ is an 8-bit integer between 0 and 255.
- ◆ Histogram, h_I , of I :
 - a 256-element array, h_I
 - $h_I(g)$ = number of pixels in I that have value g .
for $g = 0, 1, 2, 3, \dots, 255$

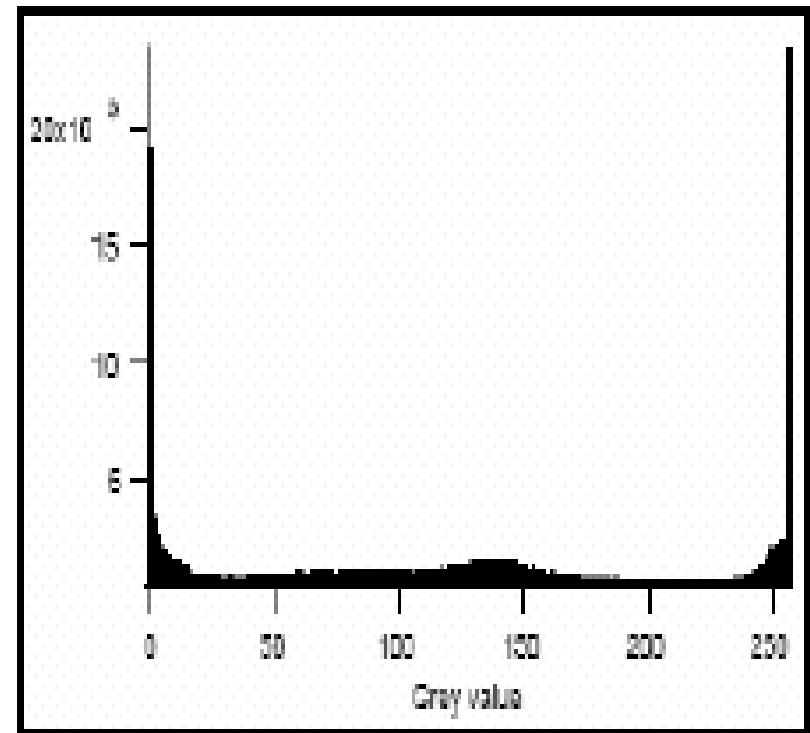
HISTOGRAM

- A discrete function $h(r_k)=n_k$
 - r_k is the k^{th} gray level
 - n_k is the number of pixels having gray level r_k in the image
- Ex:

0	1	2	3
1	3	3	0
0	1	3	0
3	0	3	1



UNIQUENESS

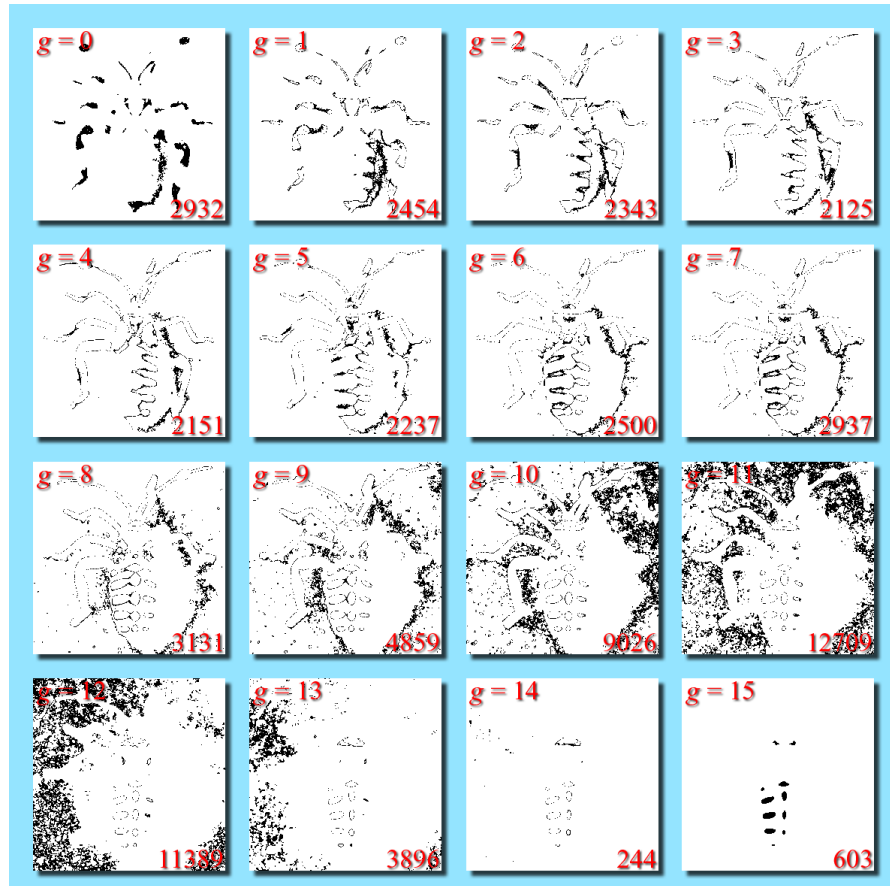


Histogram of a Grayscale Image



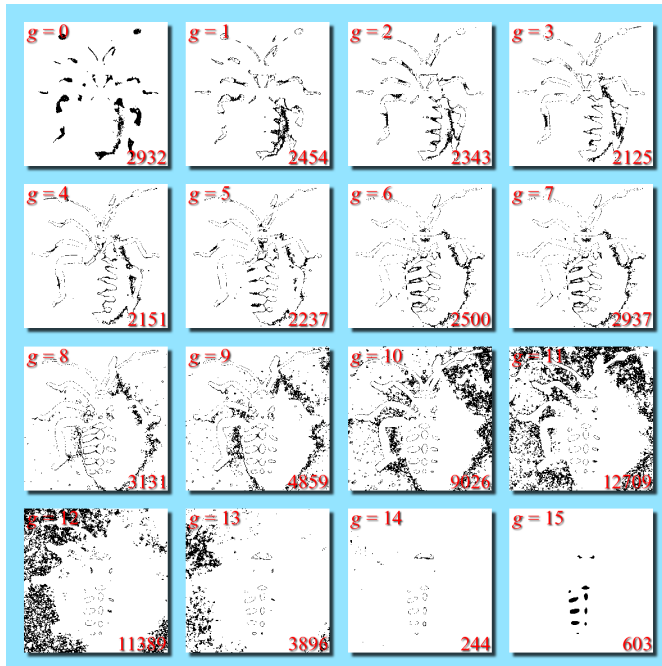
16-level (4-bit) image

lower RHC: number of pixels with intensity g



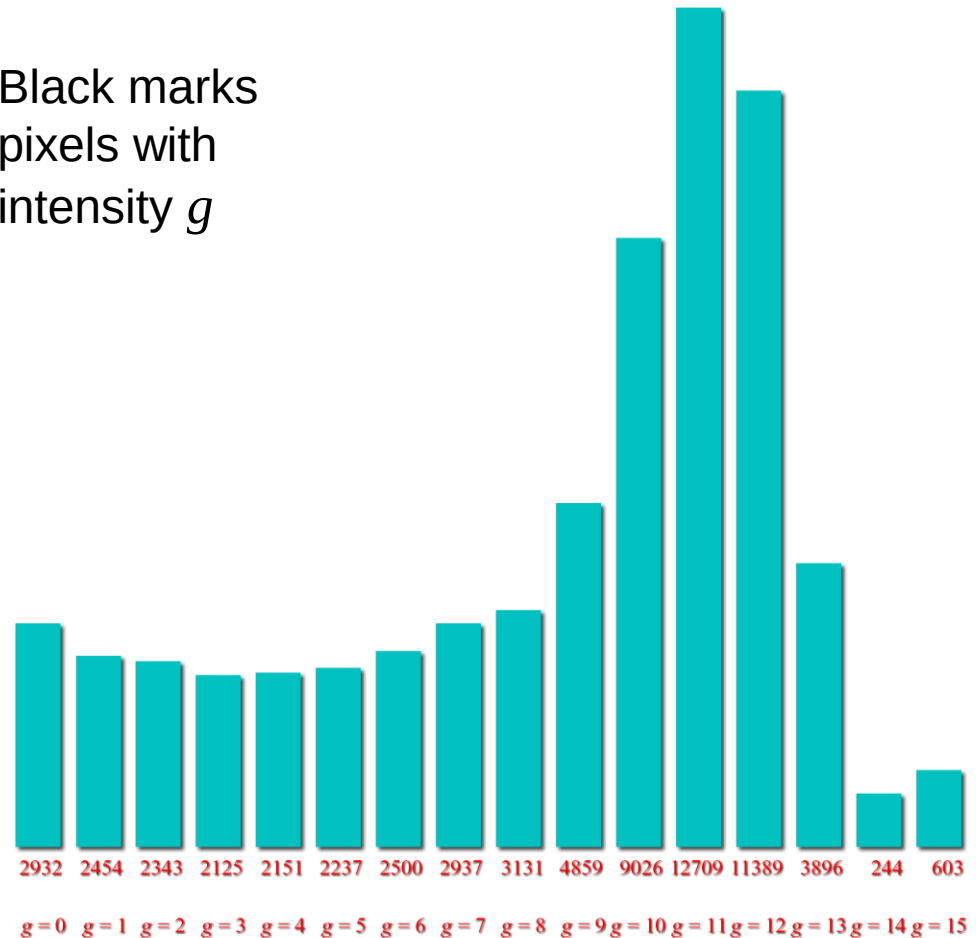
black marks pixels with intensity g

Histogram of a Grayscale Image

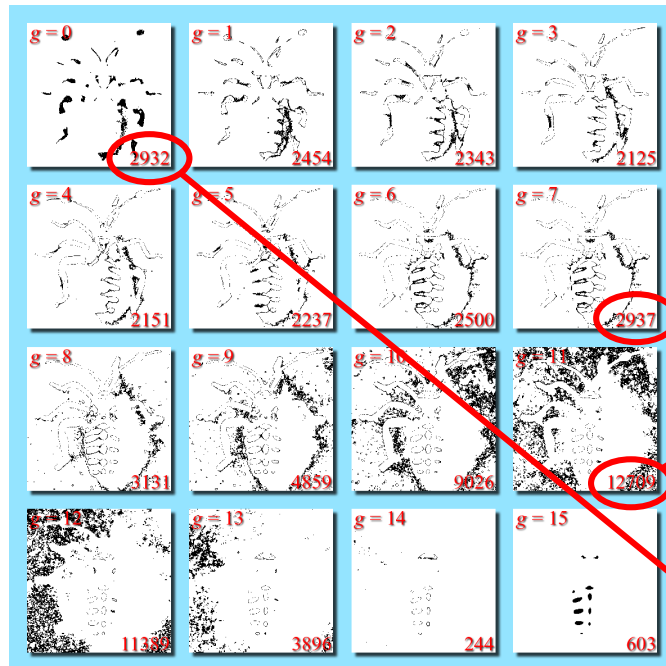


Black marks
pixels with
intensity g

Plot of histogram:
number of pixels with intensity g

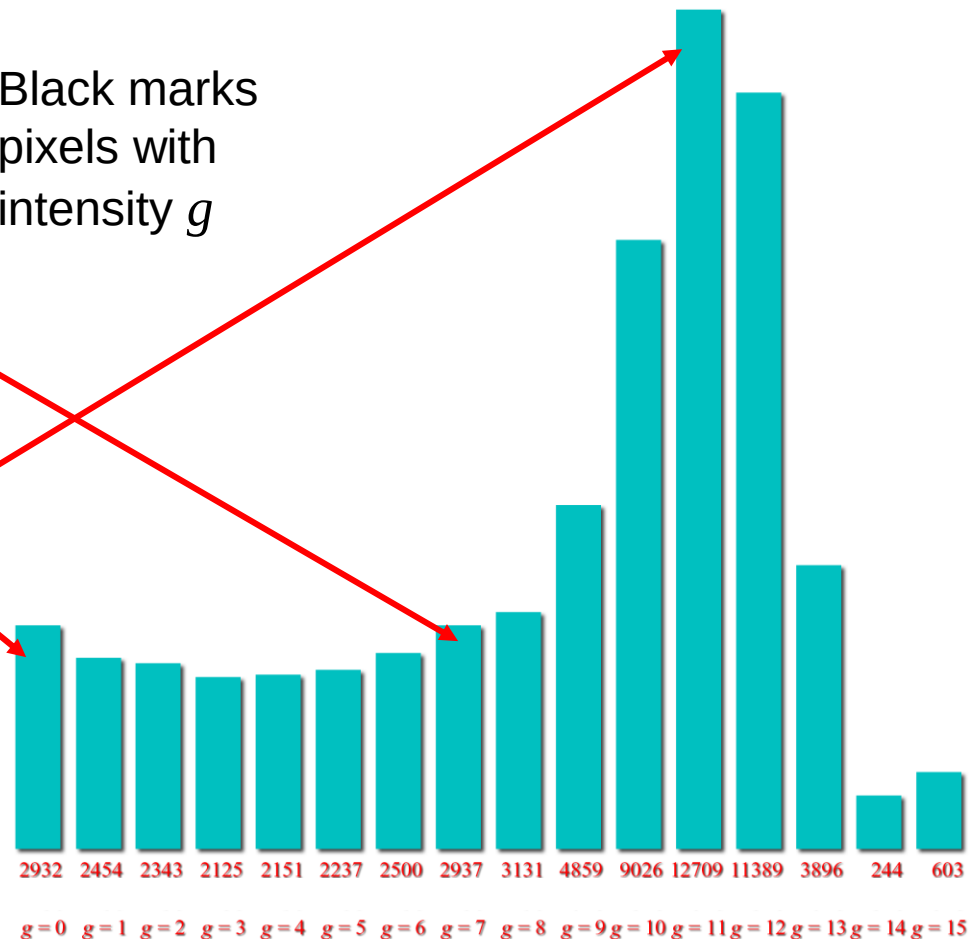


Histogram of a Grayscale Image



Black marks
pixels with
intensity g

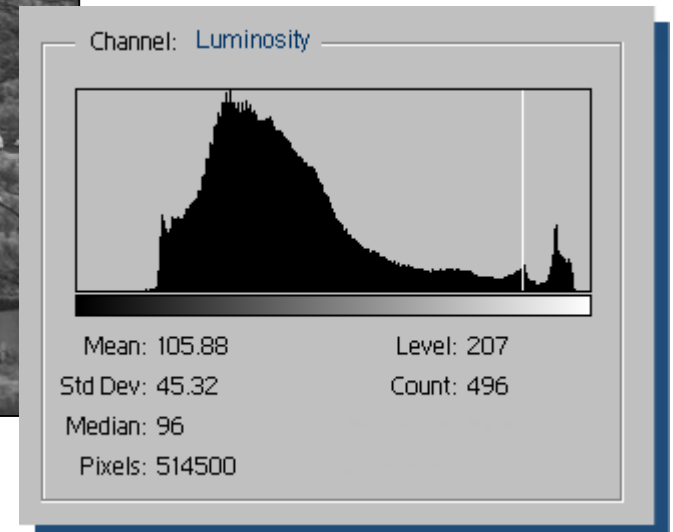
Plot of histogram:
number of pixels with intensity g



Histogram of a Grayscale Image



$h_I(g) =$ the number
of pixels in I
with graylevel g .

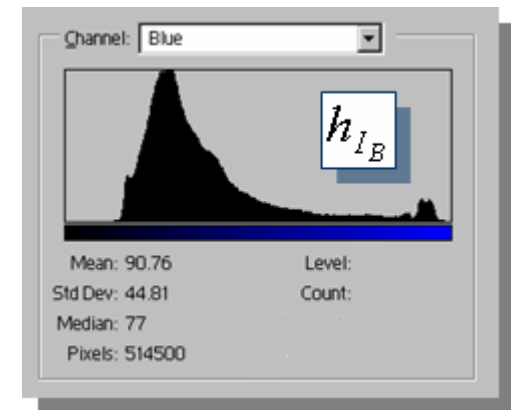
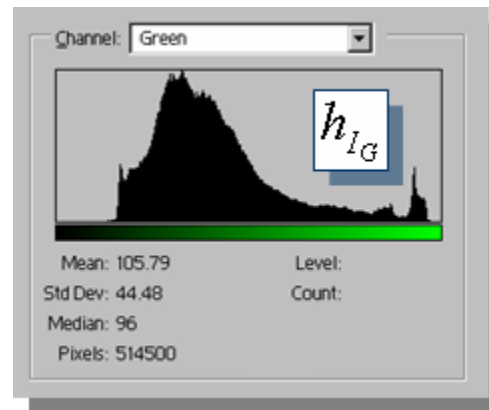
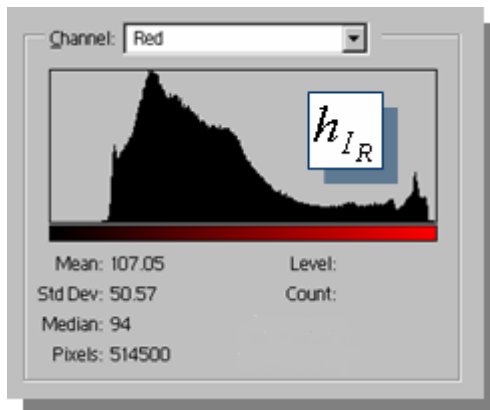
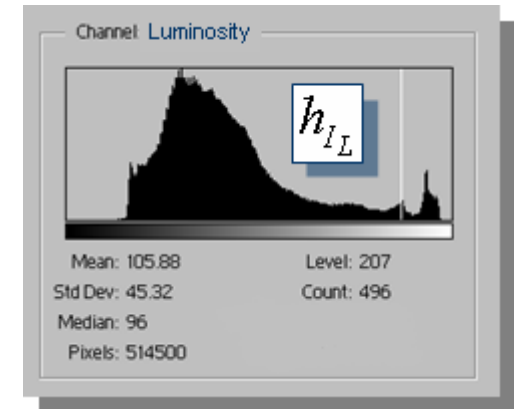


Histogram of a Color Image

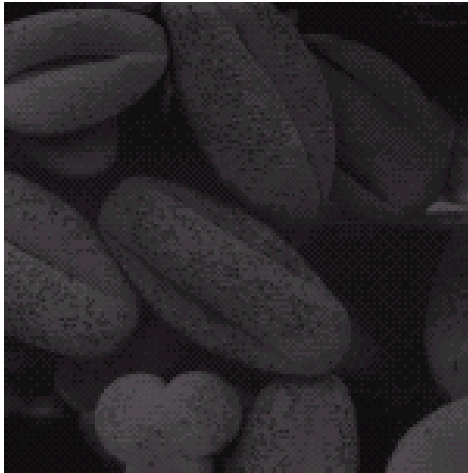
- ◆ If I is a 3-band image
- ◆ then $I(r,c,b)$ is an integer between 0 and 255.
- ◆ I has 3 histograms:
 - $h_R(g)$ = # of pixels in $I(:, :, 1)$ with intensity value g
 - $h_G(g)$ = # of pixels in $I(:, :, 2)$ with intensity value g
 - $h_B(g)$ = # of pixels in $I(:, :, 3)$ with intensity value g

Histogram of a Color Image

There is one histogram per color band R, G, & B. Luminosity histogram is from 1 band = $(R+G+B)/3$



Histogram: Example



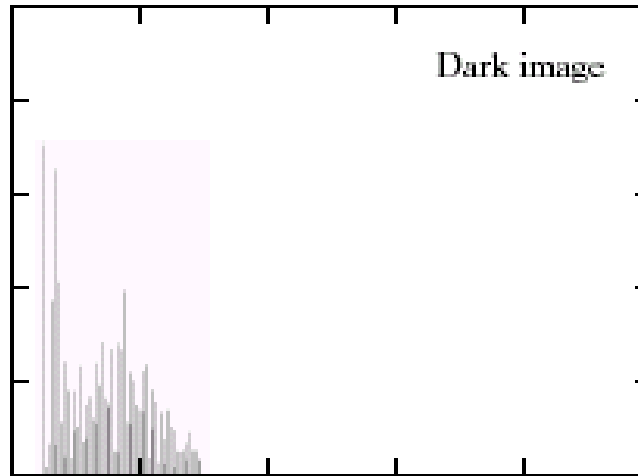
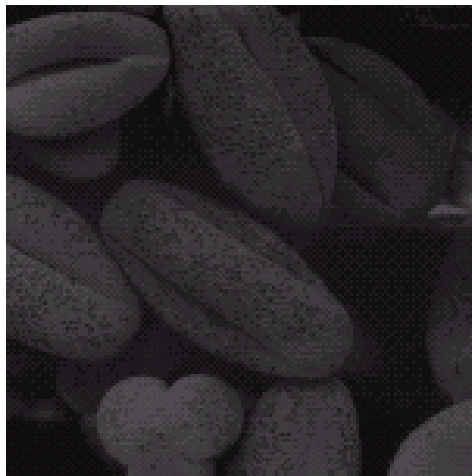
Dark Image

How would the
histograms of these
images look like?



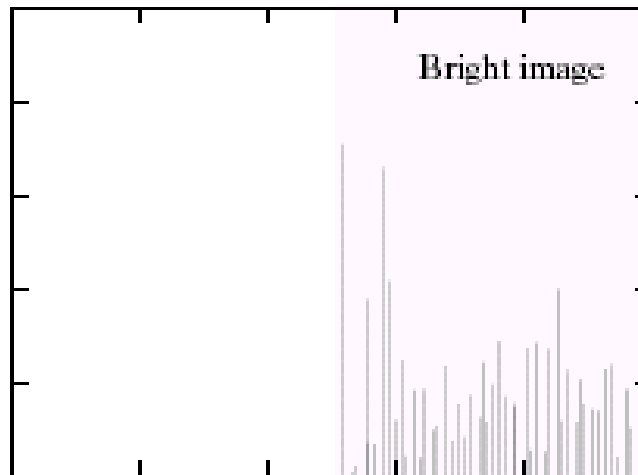
Bright Image

Histogram: Example



Dark image

Components of histogram are concentrated on the low side of the gray scale

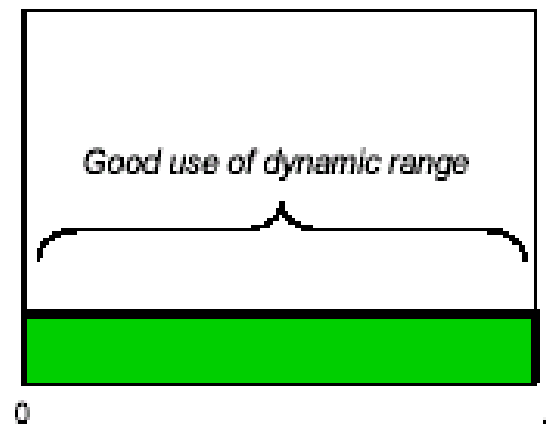
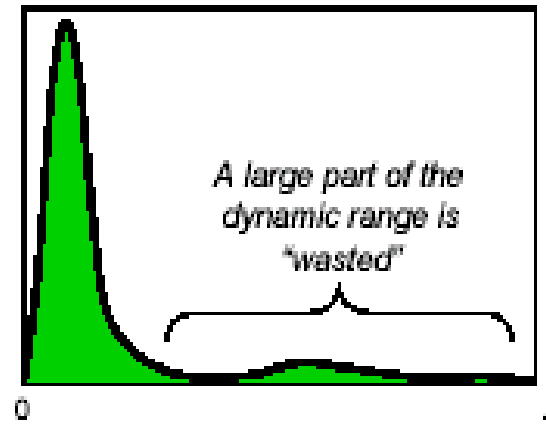


Bright image

Components of histogram are concentrated on the high side of the gray scale

HISTOGRAM INSIGHT INTO CONTRAST

- A high contrast image makes good use of the full dynamic range available.
- Hence in some applications it may be desirable to make more optimal use of the full dynamic range.
- In some circumstances this results in a clearer image.



Histogram: Example



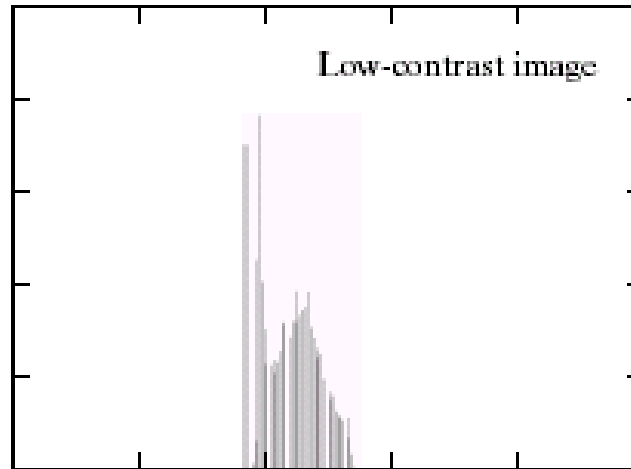
Low Contrast Image

How would the
histograms of these
images look like?



High Contrast Image

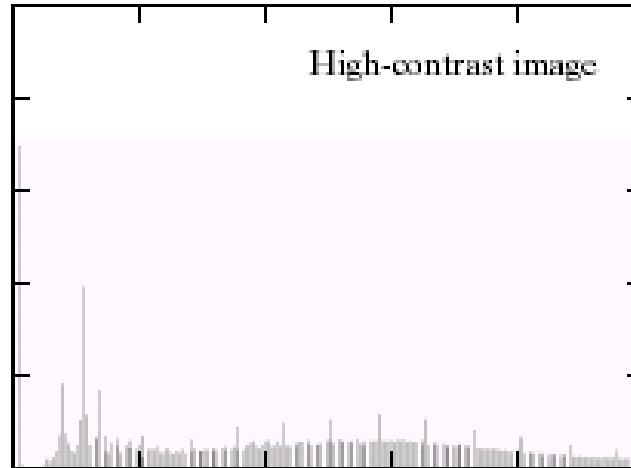
Histogram: Example



Low-contrast image

Low contrast image

Histogram is narrow and centered toward the middle of the gray scale



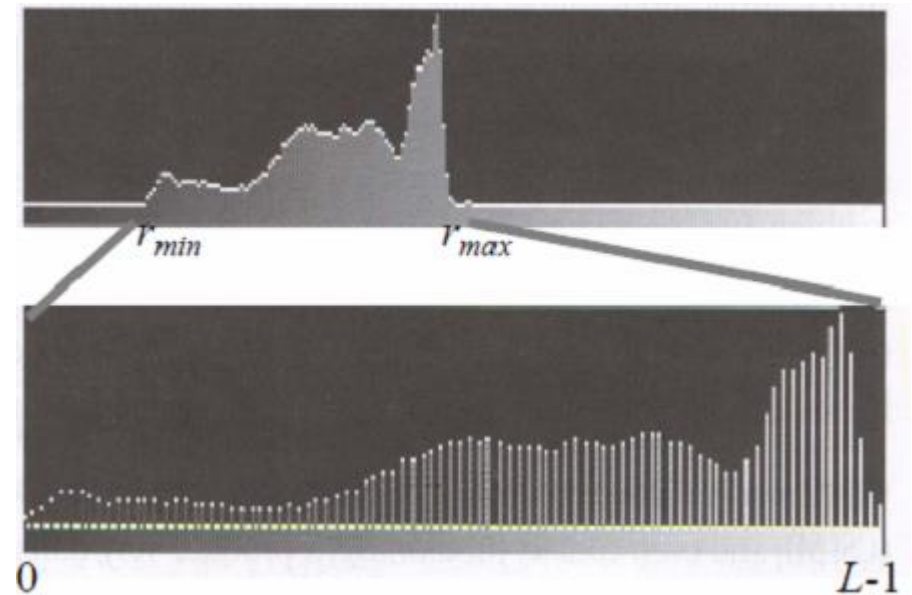
High-contrast image

High contrast image

Histogram covers broad range of the gray scale and the distribution of pixels is not too far from uniform with very few vertical lines being much higher than the others

Contrast Stretching

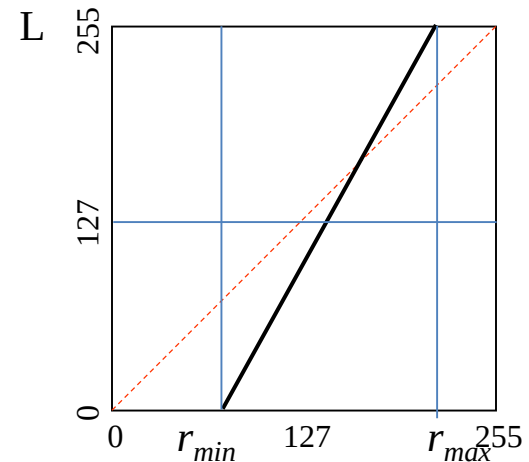
Improve the contrast in an image by 'stretching' the range of intensity values it contains to span a desired range of values, *e.g.* the full range of pixel values



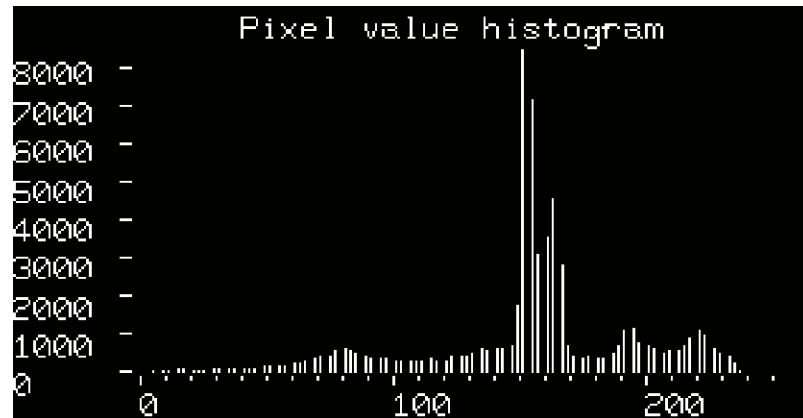
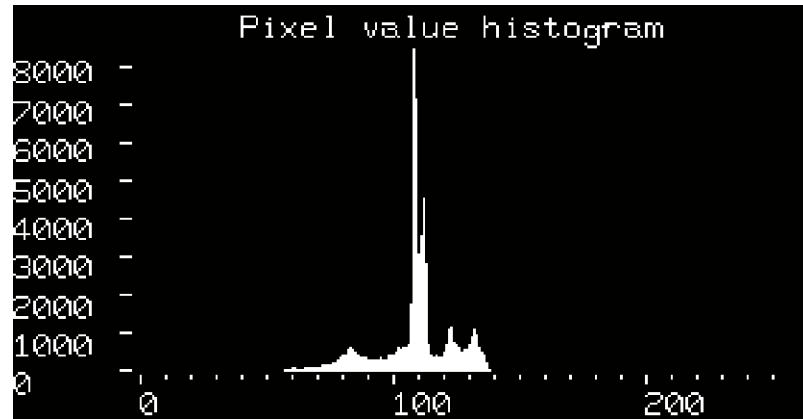
Contrast Stretching

If r_{max} and r_{min} are the maximum and minimum gray level of the input image and L is the total gray levels of output image, the transformation function for contrast stretch will be

$$s = T(r) = (r - r_{min}) \left[\frac{L-1}{r_{max} - r_{min}} \right]$$

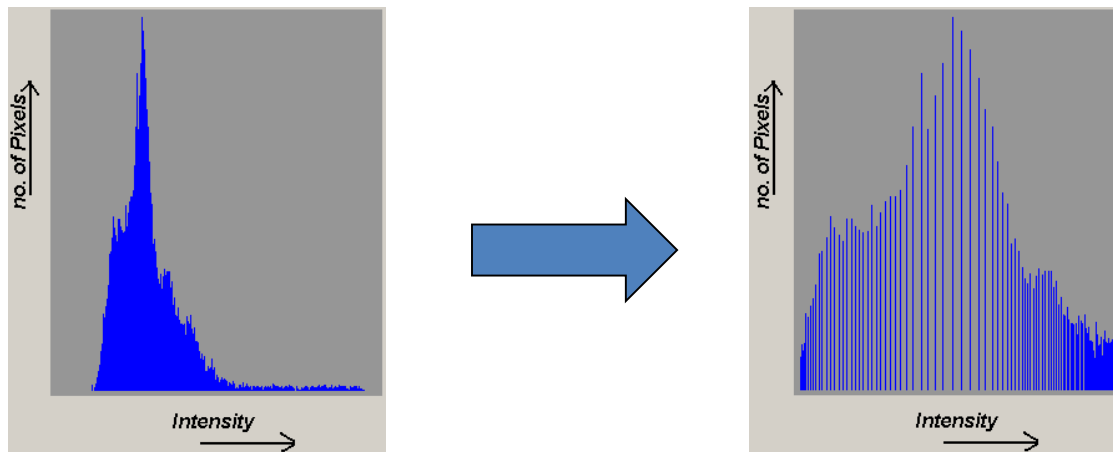


Contrast Stretching

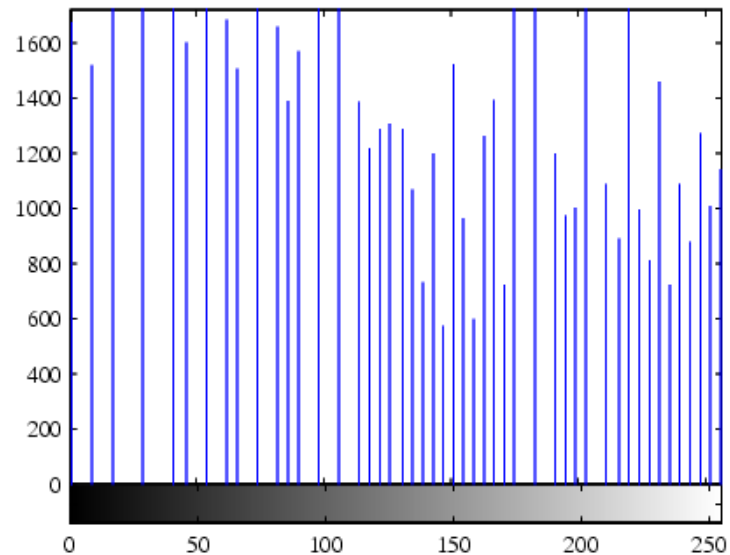
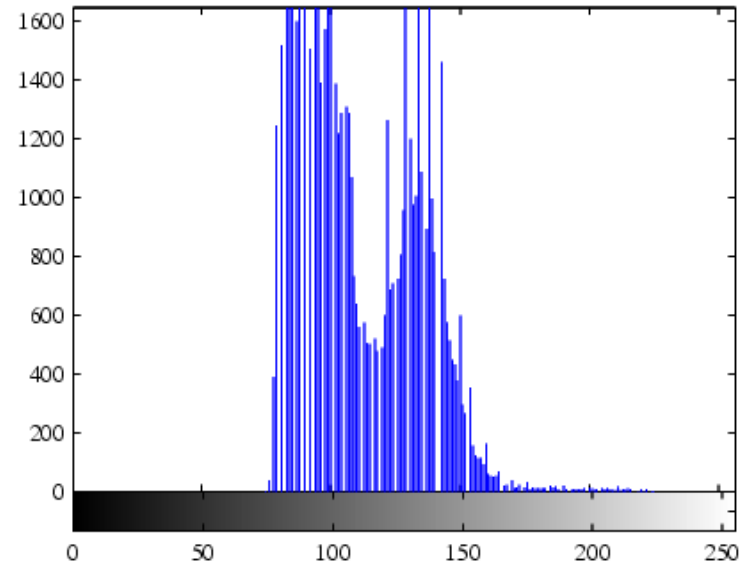


Histogram Equalization

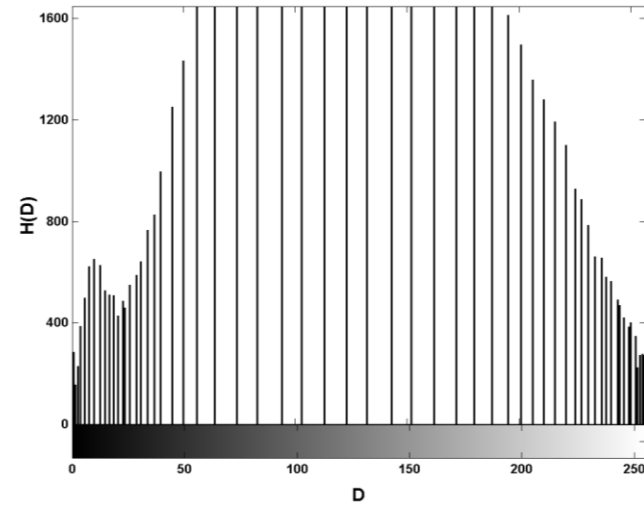
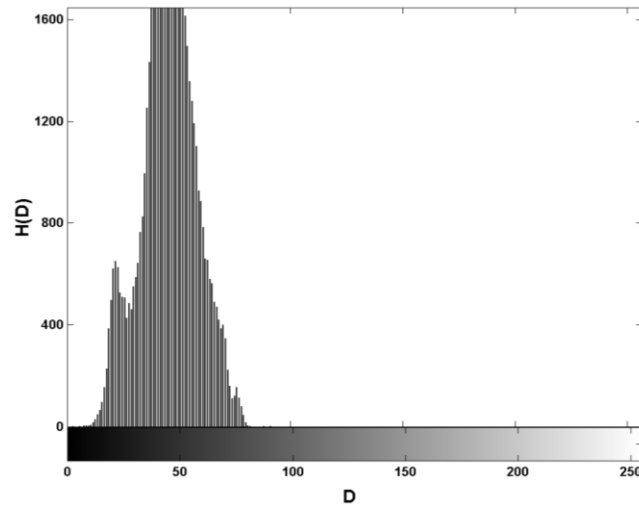
Histogram equalization re-assigns the intensity values of pixels in the input image such that the output image contains a uniform distribution of intensities



HISTOGRAM EQUALIZATION



AERIAL PHOTOGRAPH OF THE PENTAGON



**Resulting image uses more of dynamic range.
Resulting histogram almost, but not completely, flat.**

The Probability Distribution Function of an Image

$$\text{Let } A = \sum_{g=0}^{255} h_I(g)$$

Note that since $h_I(g)$ is the number of pixels in I with value g ,

A is the number of pixels in I . That is if I is R rows by C columns then $A = R \times C$.

Then,

$$p_I(g) = \frac{1}{A} h_I(g)$$

This is the probability that an arbitrary pixel from I has value g .

The Probability Distribution Function of an Image

- $p(g)$ is the fraction of pixels in an image that have intensity value g .
- $p(g)$ is the probability that a pixel randomly selected from the given image has intensity value g .
- Whereas the sum of the histogram $h(g)$ over all g from 0 to 255 is equal to the number of pixels in the image, the sum of $p(g)$ over all g is 1.
- p is the **normalized histogram** of the image

The Cumulative Distribution Function of an Image

Let $\mathbf{q} = I(r,c)$ be the value of a randomly selected pixel from I . Let g be a specific gray level. The probability that $q \leq g$ is given by

$$P_I(g) = \sum_{\gamma=0}^g p_I(\gamma) = \frac{1}{A} \sum_{\gamma=0}^g h_I(\gamma) = \frac{\sum_{\gamma=0}^g h_I(\gamma)}{\sum_{\gamma=0}^{255} h_I(\gamma)},$$

where $h_I(\gamma)$ is the histogram of image I .

This is the probability that any given pixel from I has value less than or equal to g .

The Cumulative Distribution Function of an Image

Let $\mathbf{q} = I(r,c)$ be the value of a randomly selected pixel from I . Let g be a specific gray level. The probability that $q \leq g$ is given by

$$P_I(g) = \sum_{\gamma=0}^g p_I(\gamma) = \frac{1}{A} \sum_{\gamma=0}^g h_I(\gamma) = \frac{\sum_{\gamma=0}^g h_I(\gamma)}{\sum_{\gamma=0}^{255} h_I(\gamma)},$$

where $h_I(\gamma)$ is the histogram of image I .

Also called CDF for "Cumulative Distribution Function".

This is the probability that any given pixel from I has value less than or equal to g .

The Cumulative Distribution Function of an Image

- $P(g)$ is the fraction of pixels in an image that have intensity values less than or equal to g .
- $P(g)$ is the probability that a pixel randomly selected from the given band has an intensity value less than or equal to g .
- $P(g)$ is the cumulative (or running) sum of $p(g)$ from 0 through g inclusive.
- $P(0) = p(0)$ and $P(255) = 1$;

Histogram Equalization

Task: remap image I so that its histogram is as close to constant as possible

Let $P_I(\gamma)$

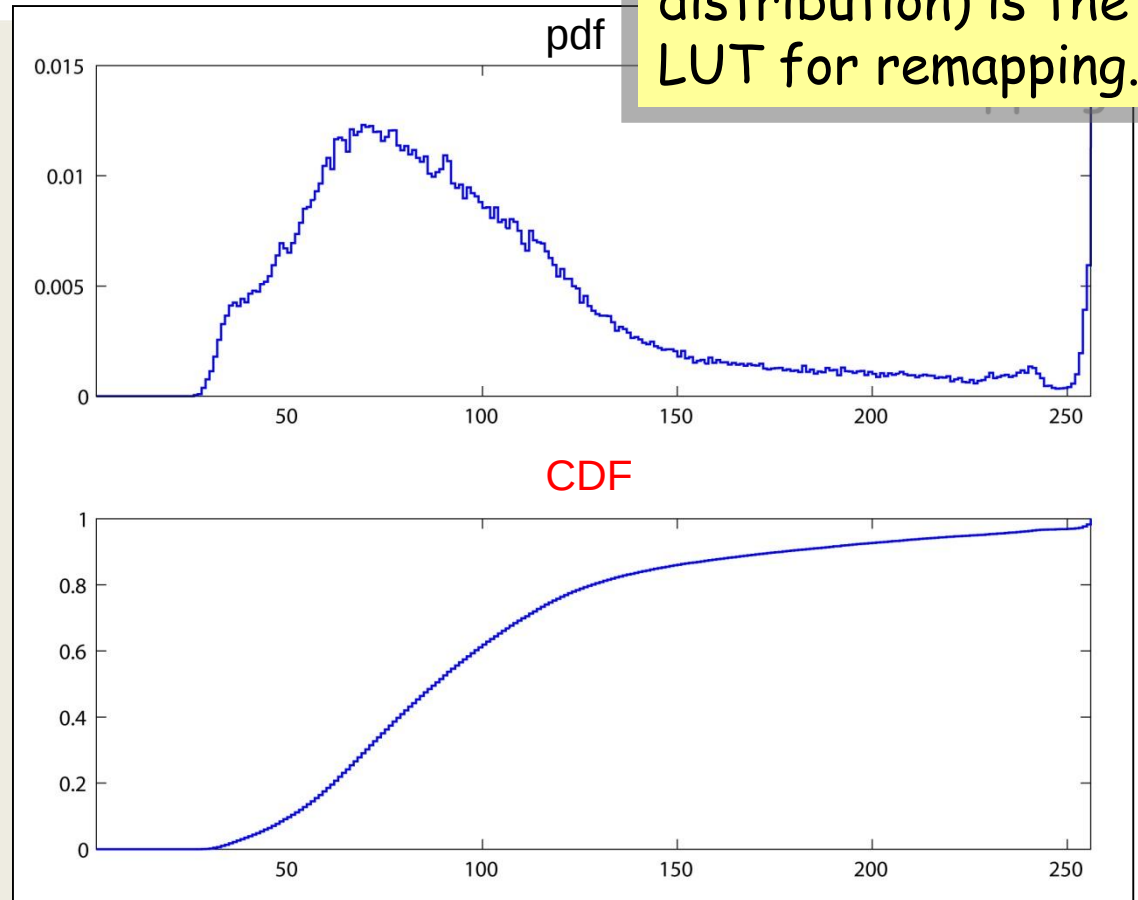
be the cumulative (probability) distribution function of I .

The CDF itself is used as the LUT.

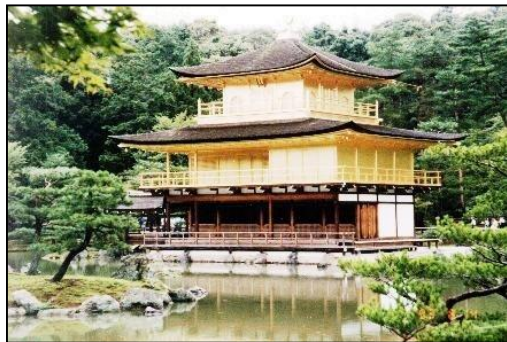
Histogram Equalization



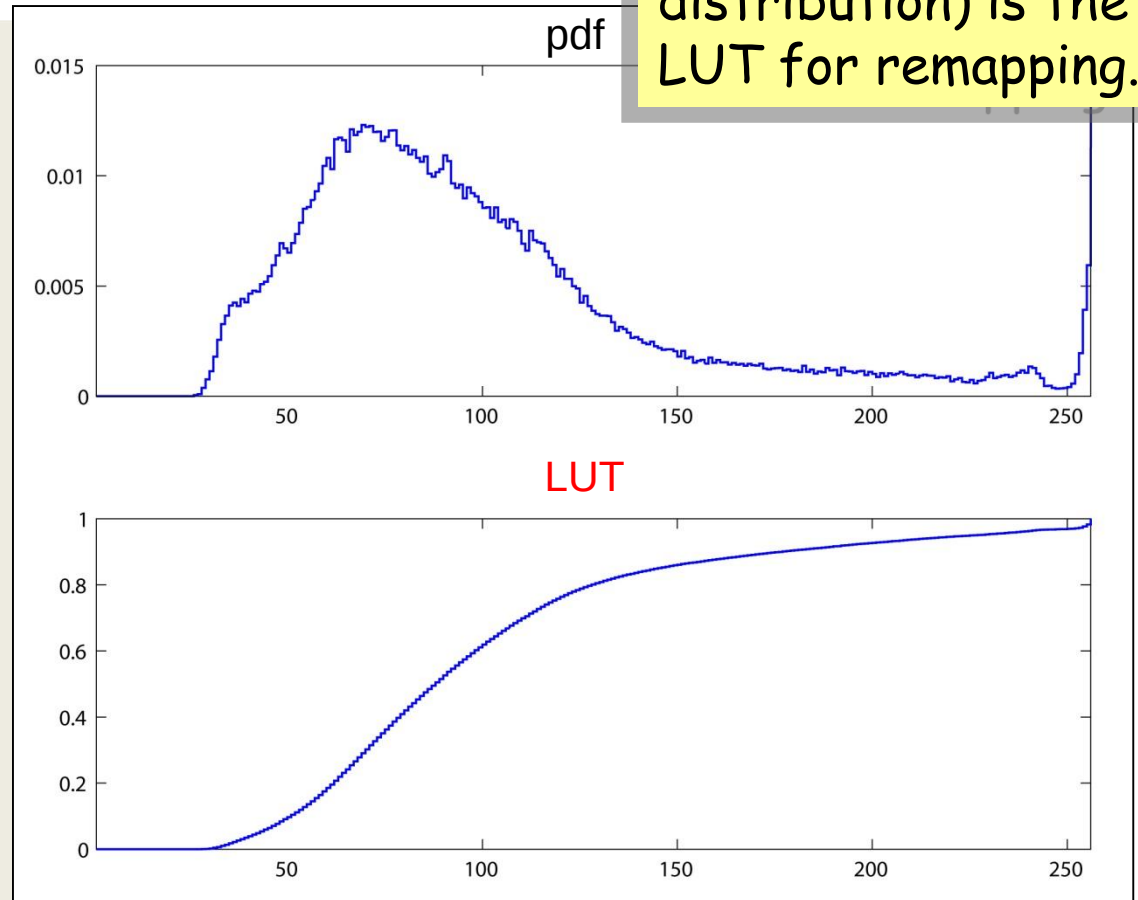
The CDF (cumulative distribution) is the LUT for remapping.



Histogram Equalization



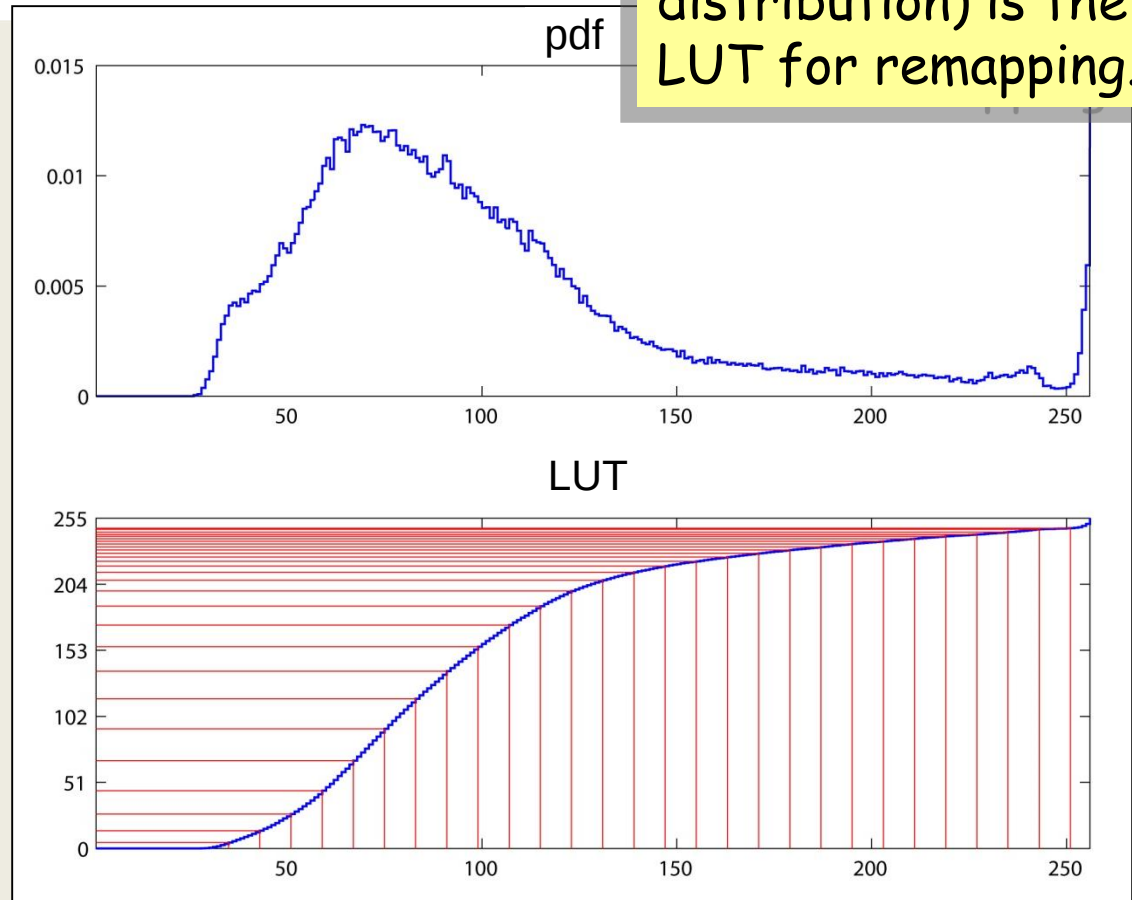
The CDF (cumulative distribution) is the LUT for remapping.



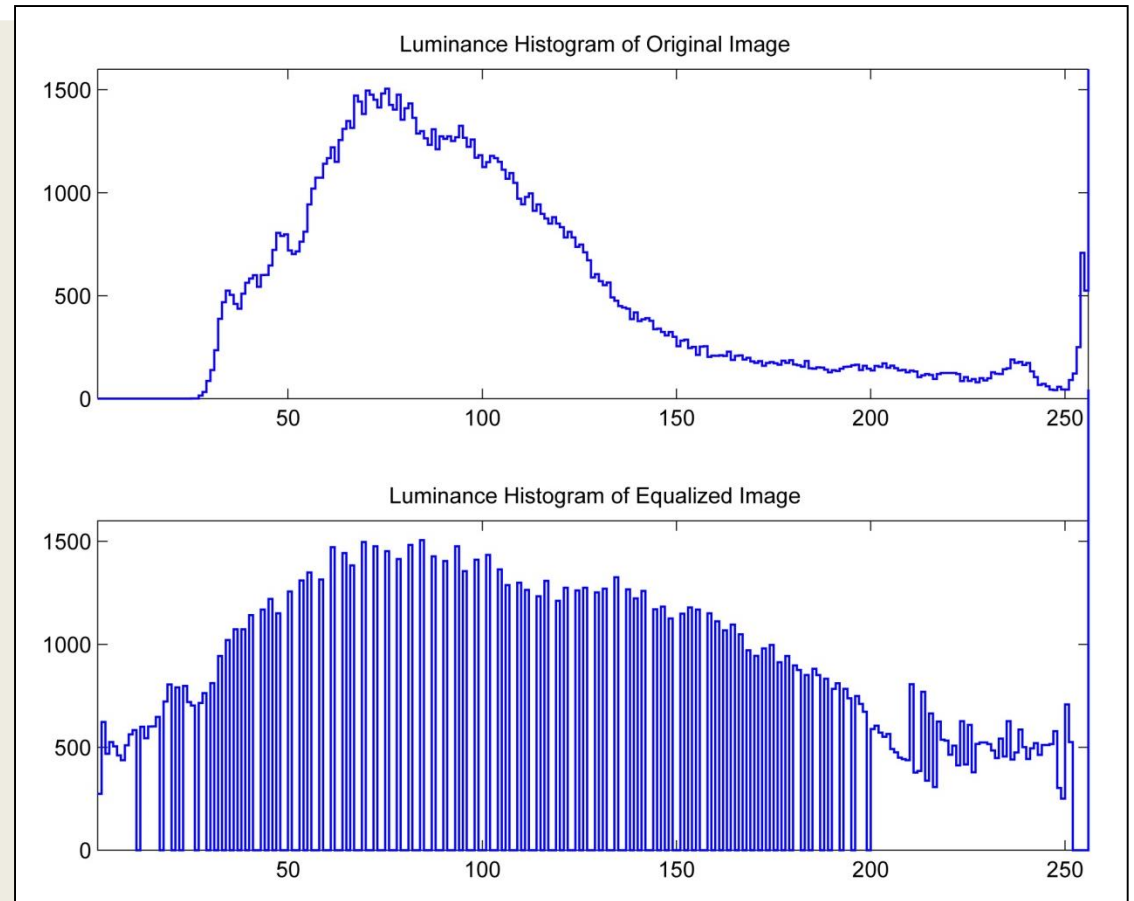
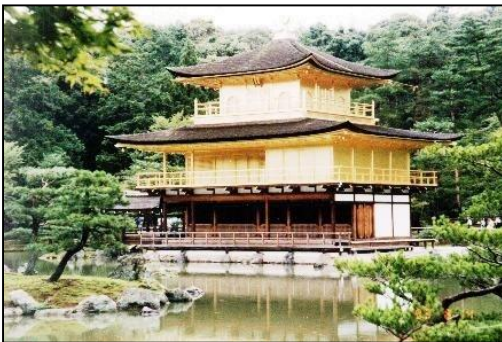
Histogram Equalization



The CDF (cumulative distribution) is the LUT for remapping.



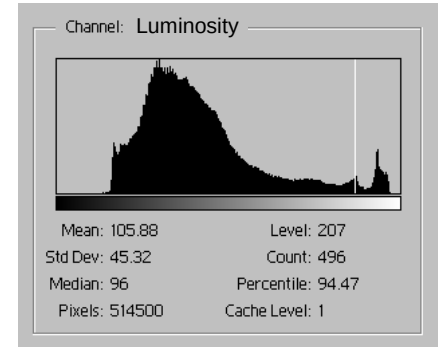
Histogram Equalization



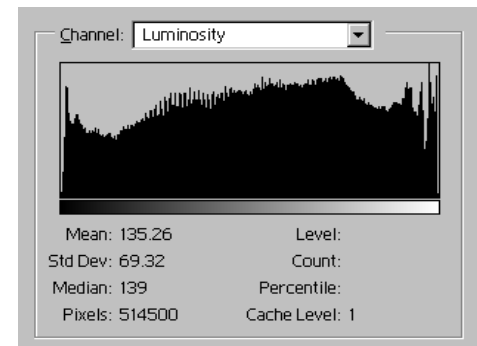
Histogram Equalization



$$J(r,c) = 255 \cdot P_I[I(r,c)].$$



before



after

HISTOGRAM EQUALIZATION IMPLEMENTATION

0	0	0	0	0
1	1	1	1	4
4	5	6	6	6
8	8	8	8	9

HISTOGRAM EQUALIZATION IMPLEMENTATION

0	0	0	0	0
1	1	1	1	4
4	5	6	6	6
8	8	8	8	9

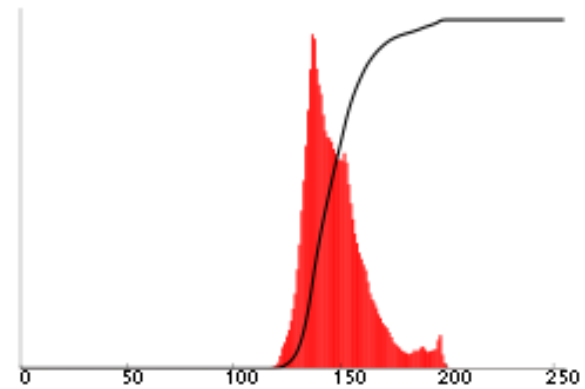
2	2	2	2	2
4	4	4	4	5
5	5	7	7	7
9	9	9	9	9

Gray levels	0	1	2	3	4	5	6	7	8	9
Counts ($h(r_k)$)	5	4	0	0	2	1	3	0	4	1
	r_0	r_1			r_2	r_3	r_4		r_5	r_6
Normalized h ($P(r_k)$)	5/20	4/20	0	0	2/20	1/20	3/20	0	4/20	1/20
cdf $F(r_k)$	5/20	9/20			11/20	12/20	15/20		19/20	20/20
$s_k = \text{round}(9 \cdot F(r_k))$	2	4			5	5	7		9	9
	s_0	s_1			s_2	s_3	s_4		s_5	s_6

Histogram Equalization: Example



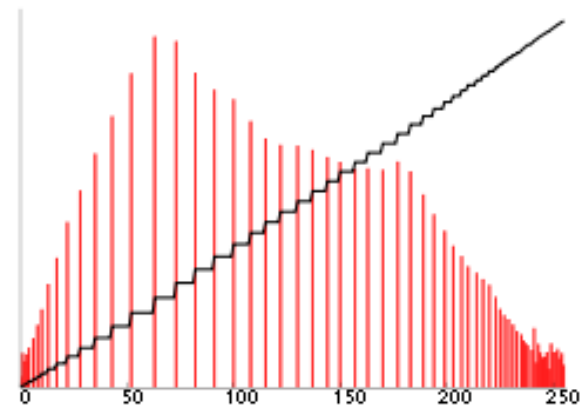
Original Image



Corresponding histogram (red) and cumulative histogram (black)



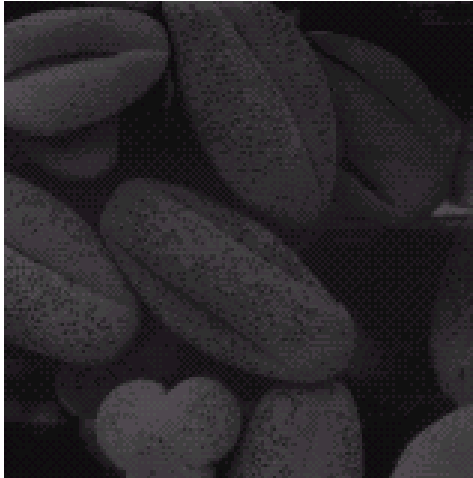
Image after histogram equalization



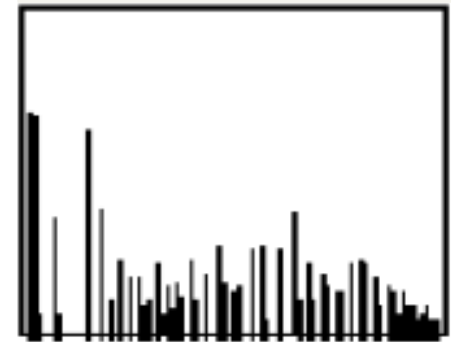
Corresponding histogram (red) and cumulative histogram (black)

Histogram Equalization: Example

Dark image



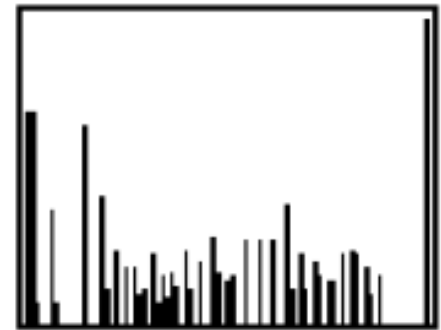
Equalized Histogram



Bright image



Equalized Histogram

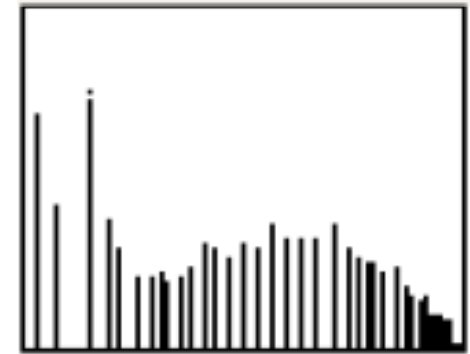


Histogram Equalization: Example

Low contrast



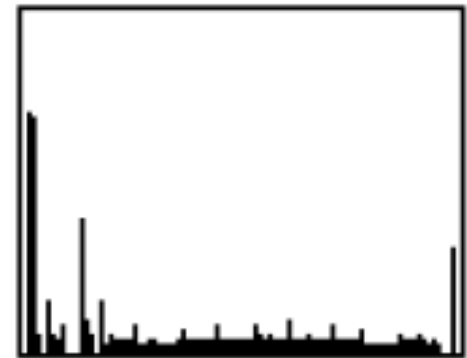
Equalized Histogram



High Contrast



Equalized Histogram



Readings from Book (3rd Edn.)

- 3.1 Image Enhancement & Transformations
- 3.2 Basic Intensity Transformation Functions
- 3.3 Histogram



Acknowledgements

- ◆ Statistical Pattern Recognition: A Review – A.K Jain et al., PAMI (22) 2000
- ◆ Pattern Recognition and Analysis Course – A.K. Jain, MSU
- ◆ *Pattern Classification*” by Duda et al., John Wiley & Sons.
- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Machine Vision: Automated Visual Inspection and Robot Vision”, David Vernon, Prentice Hall, 1991
- ◆ www.eu.aibo.com/
- ◆ Advances in Human Computer Interaction, Shane Pinder, InTech, Austria, October 2008